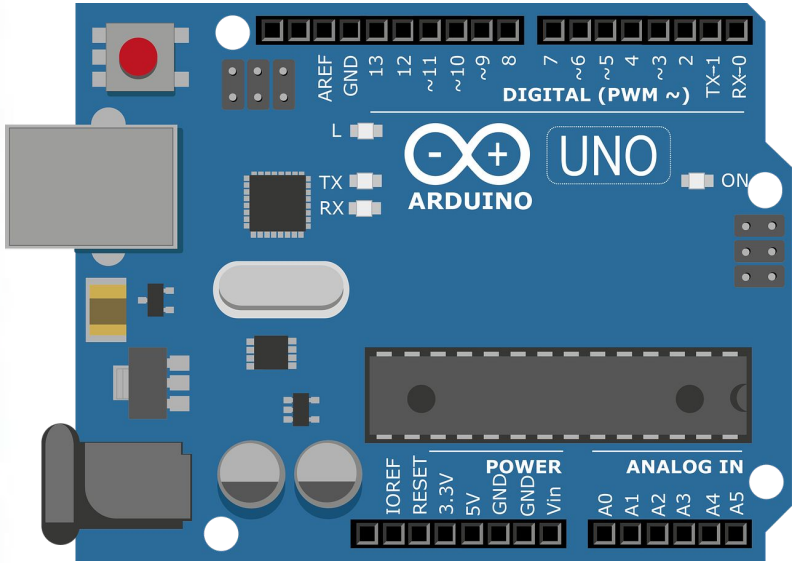


Code with Arduino

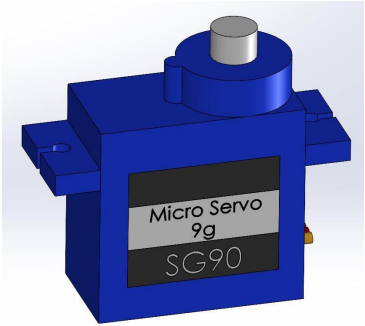
Learning how to Code a Spider

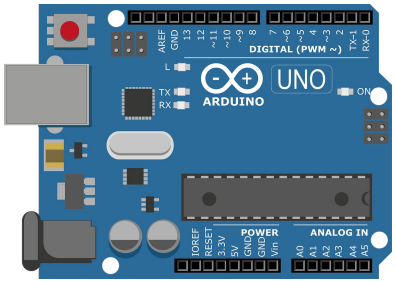
Spider

For this event, we will be developing scripts together to program an arduino robotic spider to replicate certain movements in an Aliexpress robot buy.

SG90 Robotic Spider with MG90S Kit for Arduino DIY

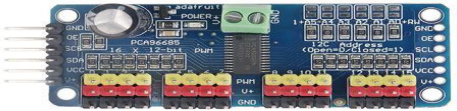
Components for this project







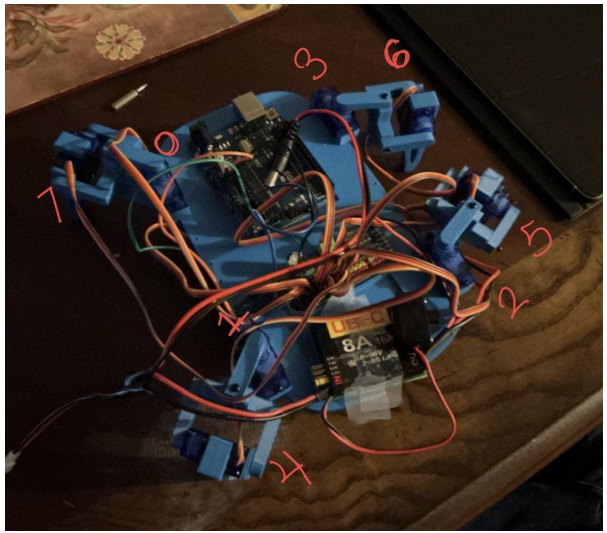




Facts about the SG90 Servos

- 180 degrees movement range
- Control - electrical pulses tell the servo which position to move too
- You must calculate the endpoints of the servo, such as 0 and 180, don’t assume that they reach the angle you want
- Calibration script to set your position

Servo Placements



Facts about the PCA9685 Servo Driver Board

- Generates electrical control signals that tell servo where to move
- 16 channels, meaning able to control 16 servos
- Uses I2C communication
- Typical movements or new cycles will occur every 20 ms
- Arduino, and board must share the same ground

Install Arduino IDE

To install:

https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE

Download the image for respective operating system

Libraries Needed

```
#include <Wire.h>
#include <Adafruit_PWMServoDriver(0x40)>
```

https://github.com/adafruit/Adafruit-PWM-Servo-Driver-Library

```
//default address is 0x40
Adafruit_PWMServoDriver board1 = Adafruit_PWMServoDriver(0x40);
```

Important Global Variables

```
const uint8_t SERVO_CHANNEL = 0;
const int PWM_FREQUENCY = 50;

// Counts at 50 Hz:
// 205 ≈ 1001 us
// 307 ≈ 1499 us
// 410 ≈ 2002 us
const int SERVOMIN = 205;
const int SERVOMAX = 410;

// Larger number = slower commanded movement.
const int STEP_DELAY_MS = 40;
```

For each script, you must state how many servos you are using, whether stating it as an individual, or stating it in an array.

PWM_Frequency to set the Hz, which according to the Github, servos run at 50 Hz updates

Servo min = minimum pulse length count
Servo max = maximum pulse length count

Step_delay = the speed of the commands

Important Functions

```
int angleToPulse(int angle) {
  angle = constrain(angle, 0, 180);
  return map(angle, 0, 180, SERVOMIN, SERVOMAX);
}
```

Function that takes in an angle position, like 90, and checks that the angle is not out of bounds from 0 to 180. We are returning a pulse count for the PCA9685 server board, which will tell the servo how much movement to perform. KEEP this in mind.

Logic Behind Rising Up

Looking at the image, servos 4-7, will move from 0 degrees to 90 degrees. Because when rising up you have to take into account the weight that the spider is experiencing. It is best to have the final position to be 70, to really push the servos into a certain position.

Another way of rising is to move one leg at a time, so start all servos at 0 and then one at a time move to 70~90 degrees.

How can we program this?

Logic Behind Waving

An important note: Know what positions you calibrated for, because then developing scripts for certain movements are easier.

Wave one “hand”, bring the leg up to 0, move respected joint to move side to side.

Applications

Use an ESP32 for a wifi html page

Add an additional servo - for this must account into calculations

Include an additional electrical component

Testing

The biggest issue was definitely the prints overall. Mainly due to the sg90 servos and attempting for them to control the joint and leg movements. Always calibrate the servos before screwing them in.

Github will be updated by the end of Tuesday and before the build event.

Questions

Thank You